ACC250: Intro to Financial Accounting

Ch6. Merchandising Operations & Multistep Income Statements

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Learning Objectives

- Apply the **inventory equation** linking **inventory** and **cost of goods sold**.
- Compare the **periodic** and **perpetual** inventory systems.
- Record purchases and sales, including **discounts**, **returns**, and **transportation costs**.
- Prepare a **multistep income statement** and compute the **gross profit ratio**.

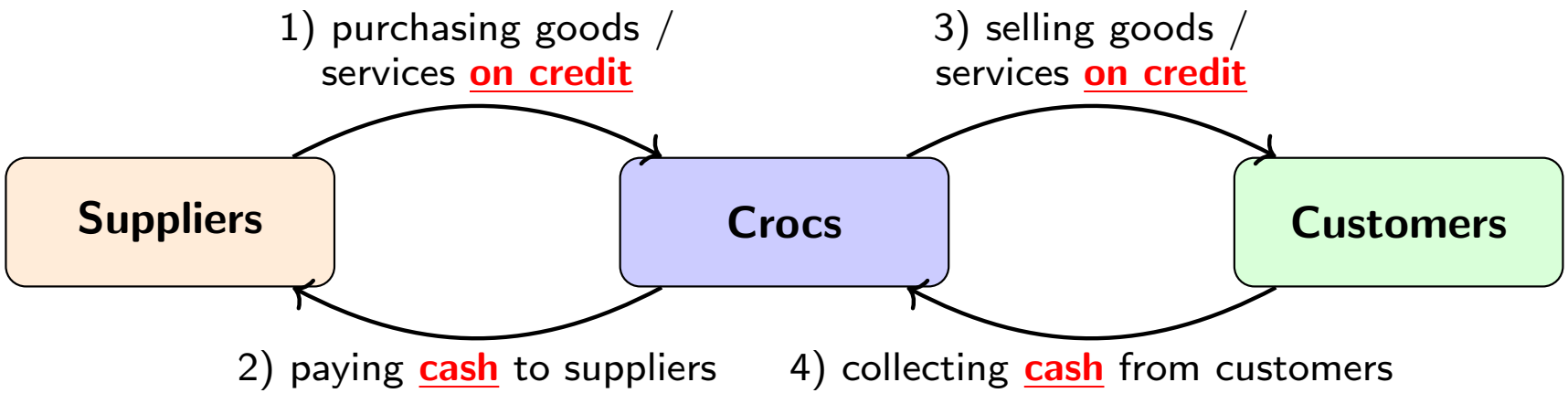
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   - Inventory Selling Transactions

3. Multistep Income Statements and Profit Ratios

Operating Cycle



Operating cycle:

- the period from the purchase of goods and services to the collection of cash from customers.[1]

[1] ACC250 covers **merchandising** firms only (e.g., Amazon, Walmart, Target): they buy finished goods and resell them. Manufacturing firms (e.g., Apple, Tesla, GM) are covered in ACC255.

Inventory - Concept

Inventory (Asset)

The total value of the goods held for sale in the ordinary course of business.

- Inventory is a **current asset**.
- Inventory is recorded at its **cost** (under the **cost principle**).
  (i.e., how much the company paid to acquire the inventory.)

Cost Principle

Assets are initially recorded at their **acquisition cost**, which includes all costs necessary to acquire the asset and prepare it for its intended use.

Inventory Calculation Example

EX. How much inventory was sold (i.e., **Cost of Goods Sold, COGS**) given the following information?

- At the end of the last year, the inventory balance was $100.
- In January, your company purchased $600 additional inventories (i.e., Purchases).
- At the end of January, you found that the inventory balance (i.e., after selling some of it) is $300.

Answer:

- **Goods available for sale**: $100 + $600 = $700
- **Cost of Goods Sold**: $700 - $300 = $400

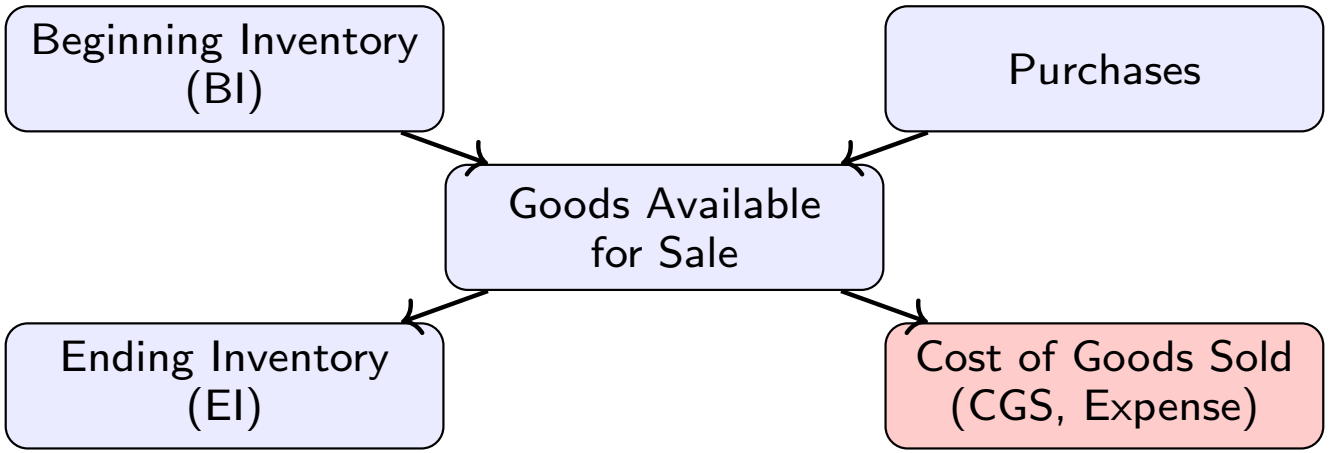
Inventory - Equation

Inventory Equation

End. Inventory = Beg. Inventory + Purchases - Cost of Goods Sold

Inventory is

- increased when purchased;
- decreased when sold.



Inventory Equation - to get EI

EX 1. Suppose a company has the following information for the year:

- Beginning Inventory (BI): \$2K
- Purchases: \$8,500
- Cost of Goods Sold: \$8,200

What is the Ending Inventory (EI) for the year?

- **Goods Available for Sale**: \$2K + \$8,500 = \$10,500
- **Ending Inventory**: \$10,500 - \$8,200 = \$2,300

Inventory Equation - to get CGS

EX 2. Suppose a company has the following information for the year:

- Beginning Inventory (BI): \$2K
- Purchases: \$8,500
- Ending Inventory (EI): \$2,300

What is the Cost of Goods Sold (CGS) for the year?[2]

- **Goods Available for Sale**: \$2K + \$8,500 = \$10,500
- **Cost of Goods Sold**: \$10,500 - \$2,300 = \$8,200

[2] This calculation is required under Periodic Inventory System.

Inventory vs Supplies

Similarities:

- Both Inventory and Supplies are **assets**.
- Both are **current assets**.
- Both are purchased from **suppliers**.
- Both are usually recorded with **Accounts Payable**.

Difference:

- **Inventory** is the goods that a company has purchased but not yet **sold**.
- **Supplies** are the goods that a company has purchased but not yet **used**.

Accounting for Inventory (with Cost of Goods Sold)

Inventory (Asset)

The total value of the goods held for sale in the ordinary course of business.

Cost of Goods Sold (Expense)

The cost of the goods sold during a period.

When inventory is purchased:

1. Inventory is recorded as an asset.
2. Cash should be paid now or in the future.

EX. The company purchased $50 inventory on account:

Inventory (+A)	50	
Accounts Payable (+L)		50

Accounting for Inventory (with Cost of Goods Sold)

When the inventory is sold:

1. Revenues are recorded.
2. Cash or Accounts Receivable is recorded.
3. Inventory is decreased.
4. Cost of Goods Sold is recorded (Matching Principle).

EX. The company sold the inventory for $100 (, purchased for $50) on account:

Accounts Receivable (+A)	100	
Revenues (+R, +SE)		100

Cost of Goods Sold (+E, -SE)	50	
Inventory (-A)		50

The two sets of entries are recorded when the inventory is sold!

Relationship between Assets and Expenses

Assets vs Expenses:

- Assets are resources that a company owns or controls.
- Expenses are resources that a company has used up to generate revenues.

Assets are turned into expenses when they are used up to generate revenues.

- **Inventory** is turned into **Cost of Goods Sold** when it is sold.
- **Supplies** are turned into **Supplies Expense** when they are used.
- **Equipment** is turned into **Depreciation Expense** when it is used.
- **Prepaid Rent** is turned into **Rent Expense** when it is used.

Accounts Payable Terms (e.g., 2/30, n/60)

Inventory and Accounts Payable:

- Inventory is usually purchased on credit.
- Inventory purchases increase **Accounts Payable**.

Contractual terms for Accounts Payable (e.g., "**2/30, n/60**" or "**2/30, g/60**"):

- **2/30**: the buyer can get a **2%** discount if they pay within **30** days.
- **60**: the buyer should pay the full amount within **60** days without any discount.

Note: The disctinction between "n" vs "g" goes beyond the scope of this course. "n" means "net" method; "g" means "gross" method. **Net method** means that the buyer should pay the full amount without any discount. **Gross method** means that the buyer should pay the full amount with the discount.

Example

EX 1. A company purchases \$5K of inventory on account from a supplier with terms **2/30, n/60**. If the company pays the invoice within 30 days, how much should be paid?

\$4,900 (after 2% discount)

EX 2. What if the company pays the invoice 45 days after the purchase?

\$5K (no discount)

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Inventory System

1. **Periodic inventory system**: Inventory is physically counted only at the end of reporting period.[3]
2. **Perpetual inventory system**: Inventory is updated perpetually (i.e., every time inventory is bought, sold, or returned).
   $\rightarrow$ More accurate and modern when automized inventory tracking is available.

[3] This system seems easy to use, but journal entries are more complex. We do not cover journal entries for this system in this course.

Periodic Inventory System: Example

Under the periodic system, **CGS** is **inferred**: **CGS** = **BI** + **Purchases** − **EI**.

EX 1. Our company counts inventory at the end of each month. The following information is available:

- Beginning Inventory (BI): $100
- Purchases: $200
- Ending Inventory (EI): $150

What is the Cost of Goods Sold (CGS) for the month?

- CGS: **$150**

Perpetual Inventory System

Inventory is updated perpetually (i.e., every time inventory is bought, sold, or returned).

- Every moment, we know **CGS** and **Inventory**.
- Inventory equation isn't needed to calculate **CGS**.

EX 1. BI= \$100. Consider the following transactions. What is the inventory balance at the end of each month? and what is the CGS? (\$ amounts are **acquisition costs.**)

Date	Purchases	Sales	Inventory Balance
Jan 1	\$300		\$400
Jan 10	\$100		\$500
Jan 20		\$200	**\$300**

- EI: **\$300**
- CGS: **\$200**

Perpetual Inventory System

EX 2. BI= \$500. Consider the following transactions. What is the inventory balance at the end of each month? and what is the CGS? (\$ amounts are **acquisition costs.**)

Date	Purchases	Sales	Inventory Balance
Feb 1	\$200		\$700
Feb 5		\$100	\$600
Feb 10	\$100		\$700
Feb 15		\$200	\$500
Feb 28		\$200	\$300

- EI: **300**
- CGS: **500**

Once again, under Perpeutal Inventory System,

- Inventory and CGS are updated every time inventory is bought and sold.
- Inventory equation is **not needed** to calculate the CGS.

Note: We use the Perpetual Inventory System in Ch6.

Perpetual inventory system:

- is used to demonstrate the accounting for inventory purchase and sale transactions.
- maintains the **inventory** and **CGS** on a perpetual basis.
- enables **inventory** and **CGS** to be updated every time inventory is bought or sold.
- is widely used in practice.

Note: The **periodic inventory system** requires the use of separate temporary accounts for PURCHASES, TRANSPORTATION, etc. This goes beyond the scope of this course.

Journal entries for Inventory Purchase Transactions

1. Walmart receives $10,500 of bikes purchased on account.

Inventory (+A)	10,500	
Accounts Payable (+L)		10,500

2. When buying the bikes, Walmart must pay the transportation cost of $200 to a trucker (Cost Principle).

Inventory (+A)	200	
Cash (-A)		200

3. Walmart returned some of the bikes to the supplier and received a $500 reduction in the balance owed.

Accounts Payable (-L)	500	
Inventory (-A)		500

Journal entries for Inventory Selling Transactions

4. Walmart sells four mountain bikes at a selling price of $200 per bike, for a total of $800 cash. The bikes had previously been recorded in Walmart's Inventory at a cost of $175 per bike, for a total cost of $700.

Cash (+A)	800	
Revenues (+R, +SE)		800

Cost of Goods Sold (+E, -SE)	700	
Inventory (-A)		700

After the above transactions, by how much have the following accounts changed?

Account	Balance
Cash	600
Accounts Payable	10,000
Inventory	9,500
Revenues	800
Cost of Goods Sold	700
Gross Profit	100

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Single-Step vs. Multi-Step Income Statements

ABC, Inc. — Income Statement — For the Years Ended Dec 31 *(amounts in millions)*

Single-Step

Account	2027	2026
Revenues:		
Net Sales	12,000	11,000
Expenses:		
Cost of Goods Sold	7,800	7,200
SG&A Expenses[1]	2,000	1,900
Other Expenses (Net)[2]	100	50
Income Tax Expense	630	555
Net Income	1,470	1,295

Multi-Step

Account	2027	2026
Net Sales	12,000	11,000
Cost of Goods Sold	7,800	7,200
Gross Profit	4,200	3,800
SG&A Expenses	2,000	1,900
Income from Operations	2,200	1,900
Other Expenses (Net)	100	50
Income before Income Tax	2,100	1,850
Income Tax Expense	630	555
Net Income	1,470	1,295

[1] SG&A: Selling, General, and Administrative Expenses (e.g., salaries and wages, utilities, rent).

[2] Other Expenses (Net): items outside core operations (e.g., interest expense or revenue); beyond the scope of this course.

Net Sales

- Some customers (e.g., 3%) return the products they purchased.
- Firms record the estimated returns as **Sales Returns and Allowances**.[4]
- Firm's **Net Sales** is the total sales revenue minus the sales returns and allowances.
- Income Statement usually starts with **Net Sales**, but it depends on the firm's policy.

ABC, Inc.
Net Sales Calculation
For the Years Ended Dec 31
(amounts in millions)

Account	**2027**	**2026**
Sales Revenue	12,300	11,200
Less: Sales Returns & Allowances	300	200
Net Sales	**12,000**	**11,000**
Cost of Goods Sold	7,800	7,200
Gross Profit	4,200	3,800
...	...	...
...	...	...
Net Income	1,470	1,295

[4] **Sales Returns and Allowances** is a contra account that is used to record the estimated returns of sales. This goes beyond the scope of this course. We do not cover Sales Returns and Allowances in detail and the related journal entries.

Gross Profit Ratio

Gross Profit

- A profitability measure that shows how much money is left over from revenues after accounting for the cost of goods sold.
- Calculated as: Net Sales − Cost of Goods Sold

Gross Profit Ratio (or Gross Profit Margin)

- A profitability ratio that measures how much **gross profit** is generated as a percentage of net sales.
- Calculated as: $\frac{\text{Gross Profit}}{\text{Net Sales}}$

- **2027:** $\frac{4{,}200}{12K} = 0.35$ or 35%
- **2026:** $\frac{3{,}800}{11K} \approx 0.345$ or 34.5%
- **Interpretation:** For every \$1 of sales, about **35** cents is gross profit in 2027.

Net Profit Ratio

Net Profit

- A profitability measure that shows how much money is left over from revenues after accounting for all expenses.
- Calculated as: Net Sales – All Expenses

Net Profit Ratio (or Net Profit Margin)

- A profitability ratio that measures how much **net income** is generated as a percentage of net sales.
- Calculated as: $\frac{\text{Net Income}}{\text{Net Revenues}}$

- **2027:** $\frac{1{,}470}{12K} = 0.1225$ or 12.25%
- **2026:** $\frac{1{,}295}{11K} \approx 0.1177$ or 11.77%
- **Interpretation:** For every \$1 of sales, about **12.25** cents is net profit in 2027.

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4 Appendix: FOB Shipping Terms (Not for exam)

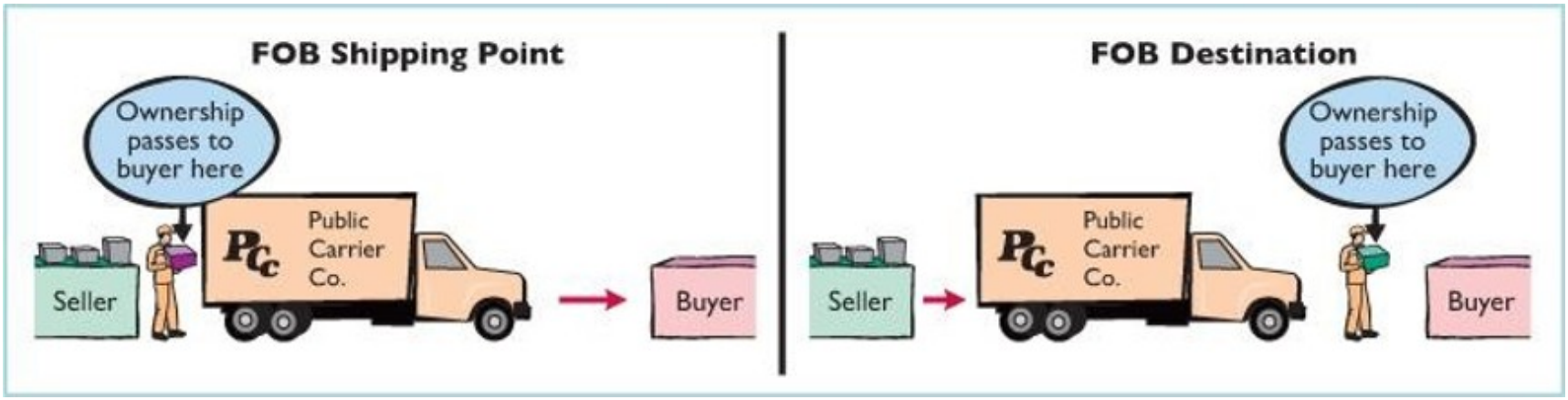
FOB shipping terms

FOB shipping point

- Ownership transfers to the buyer when goods leave the **seller's location**.
- The **buyer** is responsible for shipping costs.

FOB destination

- Ownership transfers to the buyer when goods arrive at the **buyer's location**.
- The **seller** is responsible for shipping costs.



Example - FOB Shipping Terms

EX 1. Walmart purchased $10,500 of bikes with **FOB destination**. Walmart doesn't pay the transportation cost.

Inventory (+A)	10,500	
Accounts Payable (+L)		10,500

EX 2. Walmart purchased $10,500 of bikes with **FOB shipping point**. Walmart must pay the transportation cost of $200 to a trucker (cash).

Inventory (+A)	10,500	
Accounts Payable (+L)		10,500
Inventory (+A)	200	
Cash (-A)		200